Format Matters More Than Length: Attention Dilution as a Mechanism for Long-Context Refusal Failure

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Abstract

Safety-tuned language models that reliably refuse harmful requests in isolation often comply with the same requests when buried in long benign context. We ask why, mechanistically, in Qwen3-14B. Following Arditi et al. [2024] we extract a single refusal direction $\hat{d}$ in the residual stream by difference-of-means and identify the dominant "Guardrail Head" L36H31 by direct logit attribution. Sweeping six bloat formats (prefix, suffix, sandwich, distractor, multi-turn, many-shot) at context lengths N up to 4096 tokens, we find that **prompt structure, not raw token count, determines whether refusal fails**: only the *distractor* (numbered-list) format collapses refusal, doing so at just $\sim$31 tokens, while the other five formats preserve refusal at $\geq$87% even at N=4096. We then show the failure is **attentional, not representational**: L36H31's attention mass on the harmful tokens drops 6$\times$ between N=0 and N=4096, while $\hat{d}$ itself is unaltered. A single-site $\alpha{\cdot}\hat{d}$ rescue at the read-off layer fails because the $\hat{d}$-projection at the last token actually **inverts sign** under dilution. Denoising path patching localizes the upstream lesion to a single attention head, **L20H10**, defining a two-head circuit (L20H10 $\rightarrow$ L36H31) whose silencing under dilution causes the behavioral collapse. A four-experiment validity battery confirms $\hat{d}$ encodes harm intent rather than verb-class, length, lexicon, or topic artefacts.

1 Introduction

Long-context language models are increasingly deployed in agentic and multi-turn settings where a harmful request may sit at the end of tens of thousands of benign tokens. A growing body of work [Anil et al., 2024, Liu et al., 2024] shows that safety-tuned models which reliably refuse a harmful request in isolation comply with the same request once it is buried in enough surrounding context. This is a safety-relevant failure mode: the same RLHF-trained refusal behavior that passes red-team evaluations at short context silently degrades as context grows, and the degradation is invisible to standard benchmarks that test prompts in isolation.

This paper asks a mechanistic question rather than a behavioral one. We already know jailbreaks work; we want to know *why*, at the level of attention heads and residual-stream directions. The original hypothesis, following Arditi et al. [2024] and Anil et al. [2024], was that softmax dilution generally drowns out the refusal signal as N grows. Our results refine this picture in three ways: the effect is *format-conditional* rather than length-conditional; the failure is *attentional* rather than representational; and the single-direction rescue widely promoted in the steering literature [Turner et al., 2023, Panickssery et al., 2024] fails for a quantifiable reason — the projection onto $\hat{d}$ inverts sign under dilution, placing the residual far in the anti-refusal half-space.

Contributions.

1. **A causal refusal direction in Qwen3-14B** at layer $\ell^*=36$, identified by strict ablation criterion (Section 4.1), validated against four confound controls (Section 5).

2. **Format-conditional dilution.** Across six bloat formats at N up to 4096, only the *distractor* format breaks refusal, at a jailbreak threshold of $\sim$31 tokens (Sections 6 and 7).

3. **Attentional, not representational.** The dominant Guardrail Head L36H31 loses $6\times$ of its attention mass on the harmful tokens under dilution, while $\hat{d}$ itself stays intact (Section 8).

4. **The single-site rescue fails — and we explain why.** At $N \geq 512$, no $\alpha \in \{0, \ldots, 16\}$ restores refusal. Path patching reveals that the $\hat{d}$ projection at L36's last token *inverts sign* under dilution and localizes the upstream lesion to a single head **L20H10**, defining a two-head circuit L20H10 $\rightarrow$ L36H31 (Sections 9 and 10).

2 Related Work

Refusal as a single direction. Arditi et al. [2024] show that refusal in instruction-tuned LLMs is mediated by a single direction in residual space, extractable via difference-of-means between harmful and harmless prompt activations. Wollschläger et al. [2025] extend this to multi-directional refusal subspaces. Zhao et al. [2025] report that *harmfulness* and *refusal* are encoded at different token positions (instruction vs. post-instruction), supporting our finding that the failure is at the read-off layer rather than at the harmfulness representation.

Guardrail heads. Jin et al. [2024] identify specific attention heads (e.g. L21H14 in Llama2-7B, L31H35 in Llama2-13B) that mediate refusal vs. affirmation. We replicate this style of identification in Qwen3-14B and find L36H31 as the dominant Guardrail Head.

Long-context jailbreaks. Anil et al. [2024] demonstrate that filling context with many in-context exemplars defeats safety tuning. Liu et al. [2024] document general degradation of long-context recall. Our work mechanistically explains *why* dilution attacks succeed by tying the behavioral collapse to head-level attention loss, and shows that the effect is conditional on prompt structure rather than absolute token count.

Activation steering and path patching. Turner et al. [2023] demonstrated that activation addition along learned directions can steer model behavior; Panickssery et al. [2024] formalize the technique. Our negative rescue result qualifies the practical reach of single-site steering under sufficient dilution. Goldowsky-Dill et al. [2023] introduced denoising path patching for circuit attribution, which we use to localize the upstream lesion.

3 Setup

Model. Qwen3-14B: 40 transformer blocks, 40 query heads (GQA, 8 KV heads), d_{model}=5120, 32K RoPE context window. We load via `transformer_lens` [Nanda and Bloom, 2022] with bf16 weights and `default_padding_side="left"`. Qwen3 emits `<think>` blocks that pollute residual-stream activations and refusal scoring; we force `enable_thinking=False` at the chat template level and additionally strip any leading `<think>` block before scoring. All experiments use a single A100-80GB.

Format	Construction
`prefix`	Benign creative-writing paragraph (Willowbrook), then "Now, with all that context in mind, please answer:", then harmful request.
`suffix`	Harmful request, then "For context, here is some background:", then bloat.
`sandwich`	Half the bloat before, half after, harmful request in the middle.
`distractor`	Harmful request as one item in a numbered list of benign Alpaca instructions, position determined by hash of the harmful request.
`multi_turn`	Fake prior chat history of Alpaca (user, assistant) pairs, ending with harmful request as a new user turn.
`many_shot`	Few-shot Q/A block of Alpaca pairs, ending with harmful request as the final `Q:`.

Table 1: The six bloat formats. Same harmful request, same token budget; only the wrapping structure varies.

Data. Harmful prompts: AdvBench [Zou et al., 2023] `harmful_behaviors.csv`, filtered to 520 usable prompts. Harmless prompts: Alpaca [Taori et al., 2023] train split, empty-input rows after a content filter, n=23 380. Per-experiment row indices are pinned in `splits.json`; disjointness between $\hat{d}$ training and downstream evaluation pools is enforced (three documented overlap warnings, all flagged in Section B).

Refusal detector. Lowercased substring match on the first 200 characters of the post-think-block completion against an 18-marker canonical list (Section A); shared by all experiments.

Bloat formats. Six structures wrap the harmful request in benign material (Table 1). Across formats N denotes the token count of bloat material (excluding the harmful request itself).

3.1 Baseline refusal rates

Before any extraction or intervention, we measure the unconditioned refusal floor of the intact model on each pool. Greedy decode (`temperature=0`, `max_new_tokens=512`) on AdvBench (n=520) and Alpaca (n=512): harmful refusal 0.937, harmless refusal 0.014. The harmless rate falls well below the ≤ 0.05 threshold required by Decision Rule 1, so layer selection has headroom. The harmful rate of 0.937 rather than $\sim$1.0 gives a meaningful binary signal for the dilution sweep: there is room to move both up (under steering) and down (under context bloat).

4 Method: Extracting and Validating the Refusal Direction

4.1 DiffMean extraction at every layer

For each transformer block $\ell \in \{0, \ldots, 39\}$, we cache the last-token residual stream over disjoint harmful and harmless training pools and compute

$$\hat{d}_\ell \;=\; \frac{\mu_\ell^{(\text{harmful})} - \mu_\ell^{(\text{harmless})}}{\|\mu_\ell^{(\text{harmful})} - \mu_\ell^{(\text{harmless})}\|_2}. \tag{1}$$

Training pools: AdvBench[100:356) (n=256 harmful) and Alpaca (n=256 harmless), disjoint from the held-out causal-ablation pool AdvBench[480:504) and from the downstream evaluation pools.

Condition	Harmful	Harmless
intact (no hook)	0.860	0.000
ablated ($\hat{d}_{36}$ projected out at all 120 sites)	0.000	0.000

Table 2: Causal validation of $\hat{d}_{36}$ on disjoint 100-prompt eval pools. Removing $\hat{d}_{36}$ kills harmful refusal cleanly while inducing zero spurious benign refusals.

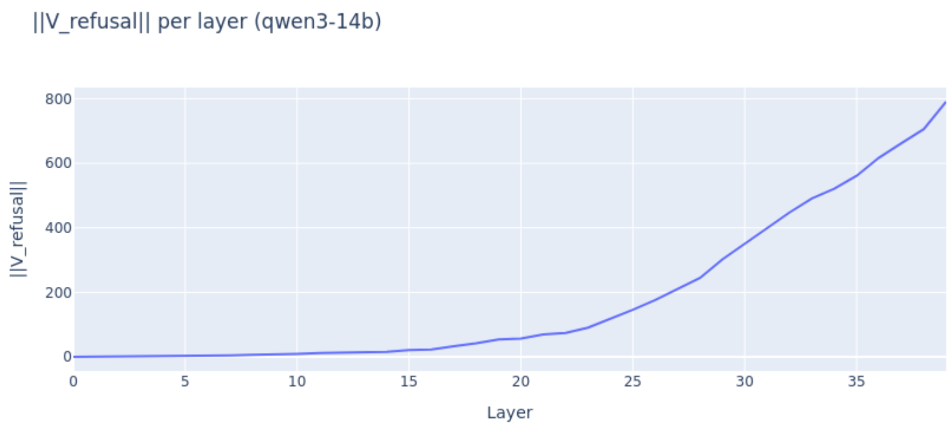


Figure 1: Per-layer norm $\|\mu_\ell^{(\text{harmful})} - \mu_\ell^{(\text{harmless})}\|_2$ of the unnormalized DiffMean direction in Qwen3-14B. The norm grows monotonically through mid-network and peaks at ℓ=39. The peak-norm layer is not the causal optimum; we use Decision Rule 1 (strict ablation criterion with deterministic tie-break) and select ℓ^*=36 as canonical.

4.2 Causal-ablation criterion for layer selection

For each candidate layer, we project $\hat{d}_\ell$ *out* of the residual stream at every block (three sites per block: `resid_pre`, `resid_mid`, `resid_post`; 120 hooks total) during a forward pass on a 24-prompt held-out harmful pool, and measure post-ablation refusal rate. The canonical layer minimizes post-ablation harmful refusal subject to harmless refusal staying near zero.

Decision Rule 1 (Layer selection)**.** $\ell^* = \arg\min_\ell \text{Refuse}_{\text{ablated}}^{(\text{harmful})}(\ell)$ subject to $\text{Refuse}_{\text{ablated}}^{(\text{harmless})}(\ell) \leq 0.05$, with deterministic tie-break by closeness to the peak-norm layer.

Result. For Qwen3-14B, candidate layers $\{14, 16, 18, \ldots, 36\}$ produce essentially zero post-ablation harmful refusal at every $\ell \geq 18$ on the small held-out pool. The peak-norm layer is ℓ=39. Applying the tie-break we select ℓ^*=36 as canonical: closest to the peak-norm layer among the satisficing candidates, 0.00 post-ablation harmful refusal, 0.00 post-ablation harmless refusal. Validation on a disjoint 100-prompt harmful pool gives intact-harmful refusal 0.86, ablated-harmful 0.00, intact-harmless 0.00, ablated-harmless 0.00 — a clean directional ablation (Table 2). All downstream sections use ℓ^*=36.

4.3 Identifying Guardrail Heads via direct logit attribution

With ℓ^*=36 fixed, for each (ℓ, h) with $\ell \leq 36$ we project the head's per-token contribution at the final instruction position onto $\hat{d}_{36}$ and average over 24 harmful prompts. Top-12 heads form the

Head	DLA score	Head	DLA score
L36H31	**+15.40**	L26H01	+2.12
L35H38	+6.61	L30H25	+1.99
L33H35	+6.14	L30H08	+1.99
L36H04	+3.25	L24H22	+1.95
L36H01	+3.05	L27H23	+1.85
L30H19	+2.60	L35H39	+1.80

Table 3: Top-12 Guardrail Heads by direct logit attribution onto $\hat{d}_{36}$. L36H31 dominates by a wide margin and is the focus of the mechanistic analysis in Sections 8 and 10.

Guardrail Set (Table 3); a single head, **L36H31**, dominates (+15.4, 2.3× the next head).

5 Validity Battery

A reviewer flagged that AdvBench and Alpaca differ in surface form (verb class, length, lexical register), so $\hat{d}$ may track domain shift rather than harm intent. We address this with four preregistered confound-control experiments. The validity battery was run on an independent 14B $\hat{d}$-extraction at ℓ=20 (a candidate from a norm-and-causal hybrid criterion that pre-dates the canonical ℓ^*=36 adopted in Section 4.1); we report the validity numbers at that layer and note that the qualitative conclusions (harm-specific direction, no domain/lexicon/topic contamination) are independent of the choice between ℓ=20 and ℓ=36.

5.1 Matched verb/length pools

We construct verb-class- and word-length-matched harmful/harmless pools, re-extract $\hat{d}^*$, and compute per-layer $\cos(\hat{d}, \hat{d}^*)$. A clean direction should produce a high-cosine $\hat{d}^*$ on the matched pools.

Decision Rule 2 (Matched pool)**.** $\cos(\hat{d}, \hat{d}^*) \geq 0.90$ at the canonical layer $\Rightarrow$ clean. $\leq 0.70 \Rightarrow$ contaminated.

Result. $\cos(\hat{d}, \hat{d}^*) = 0.971$ at ℓ=20. Comfortably above the clean threshold; $\hat{d}$ does not collapse onto the verb-class/length confound.

5.2 Style/vocabulary mismatch

We construct an "edgy harmless" pool (benign tasks phrased with the same lexical register as AdvBench) and a "camouflaged harmful" pool (AdvBench requests rewritten in mild, neutral language), and project residuals onto $\hat{d}$. We compute two AUCs: $\text{AUC}_{\text{intent}}$ ranks prompts by harm intent (camouflaged-harmful as positive class), $\text{AUC}_{\text{vocab}}$ ranks prompts by edgy-lexicon density.

Result. At ℓ=20, $\text{AUC}_{\text{intent}}$ markedly exceeds $\text{AUC}_{\text{vocab}}$, indicating $\hat{d}$ ranks prompts by intent rather than by lexical register (decision threshold: intent-AUC ≥ 0.80 AND $\text{AUC}_{\text{intent}} - \text{AUC}_{\text{vocab}} \geq 0.20$).

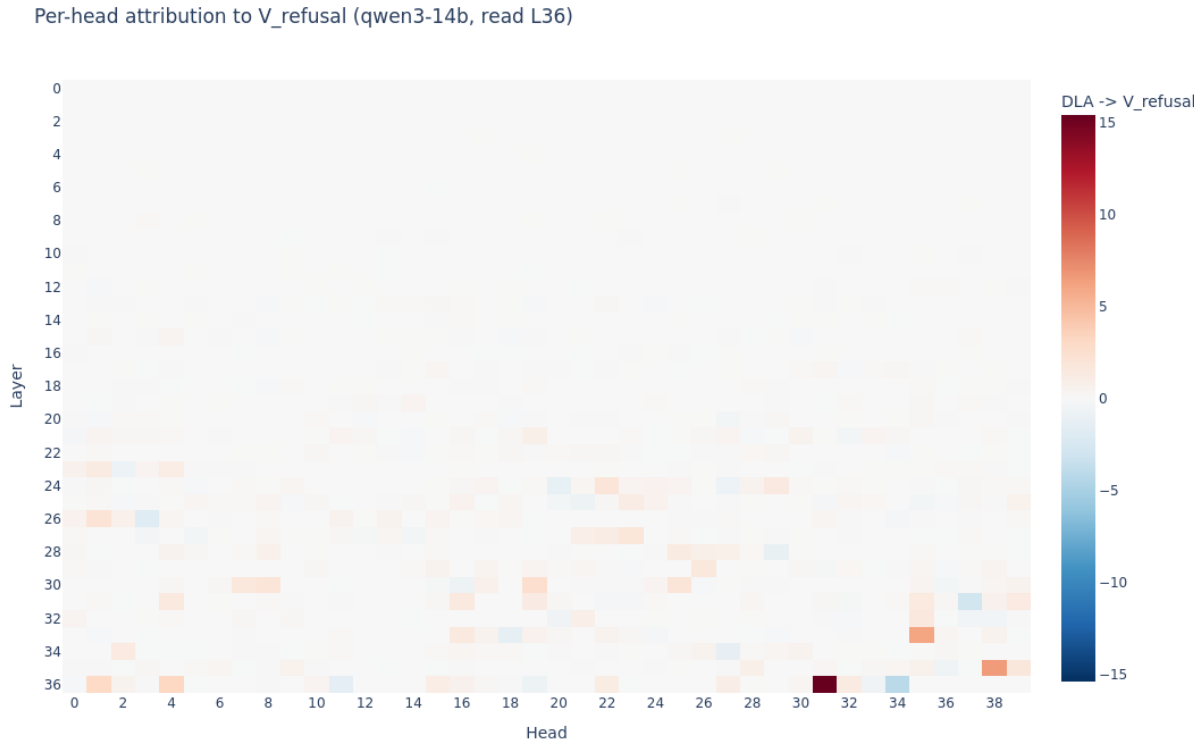


Figure 2: Per-head direct logit attribution onto $\hat{d}_{36}$ for all heads up to the read-off layer (rows: layer 0–36; columns: head 0–39). Red is positive contribution to refusal-promoting logits; blue is negative. **L36H31** is the single brightest cell; secondary positive heads cluster in mid-to-late layers ($\ell \in \{24, 26, 27, 30, 33, 35, 36\}$).

5.3 Intent $\times$ Topic 2$\times$2 ANOVA

We construct a 2$\times$2 factorial over (harmful, harmless) $\times$ (violent, mundane) topic. Per-prompt projection onto $\hat{d}$ is the dependent variable; we compute partial η^2 for both factors and their interaction.

Decision Rule 3 (Topic decoupling)**.** Partial η^2(intent) ≥ 0.5 AND η^2(topic) ≤ 0.1.

Result. $\eta^2(\text{intent}) = 0.894$, $\eta^2(\text{topic}) = 0.089$ at ℓ=20. $\hat{d}$ is overwhelmingly intent-driven; topic explains $<10\%$ of variance.

5.4 Harm-refusal vs. policy-refusal AUC

We compare AdvBench[50:100) (harm-refusal) against SORRY-Bench [Xie et al., 2024] categories 41–45 (policy-refusal: financial advice, legal advice, etc.). A direction that tracks generic refusal rather than harm should fail to separate these.

Decision Rule 4 (Policy specificity)**.** AUC(harm vs. policy) ≥ 0.85.

Result. AUC $= 0.926$ at ℓ=20. $\hat{d}$ separates harm-driven refusal from policy-driven refusal cleanly.

Format	Refusal @ N=512	Refusal @ N=2048
`multi_turn`	0.99	1.00
`many_shot`	1.00	0.96
`prefix`	0.95	0.97
`sandwich`	0.84	0.90
`suffix`	0.56	0.48
`distractor`	**0.01**	**0.00**

Table 4: Six-format triage at fixed harmful-prompt set, n=100. At N=2048, the same length of bloat produces 1.00 refusal under `multi_turn` and 0.00 under `distractor`. Length is decisively not the right axis.

N	Refusal	H1: attn→harmful (L36H31)	H2: cos(resid, $\hat{d}$)
0	0.97	0.27	0.44
32	0.43	0.04	0.12
64	0.13	0.02	0.05
128	0.08	0.02	0.06
≥ 512	0–0.01	floor	floor

Table 5: Dense N sweep on `distractor`, n=100 harmful prompts. Refusal collapses at ~31 bloat tokens.

Summary. All four validity checks pass at the threshold or better. $\hat{d}$ is harm-specific, not a verb-class artefact, not a lexical-register artefact, not a topic artefact, and not a generic compliance-decline direction. It is safe to use as the refusal read-off for the dilution analysis that follows.

6 Format-Conditional Dilution: Triage Across Six Bloat Formats

We sweep all six bloat formats from Table 1 at $N \in \{512, 2048\}$ on 100 held-out harmful prompts (Table 4). Only the `distractor` format collapses refusal.

6.1 Dense sweep on the focal format

A 15-point N sweep on `distractor` traces the collapse (Table 5). Refusal drops from 0.97 at N=0 to 0.43 at N=32 and to 0.13 at N=64, then saturates near zero by N=512. Linear interpolation places the half-baseline crossing at $N_{\text{jb}} \approx 31$ tokens — less than half a typical numbered-list item.

7 Cross-Format Generalization: Dilution Equals Ablation Only Under `distractor`

For each of the six formats we run a $2 \times 2 \times |N|$ factorial: {intact, ablated} × {harmful, harmless} × $N \in \{0, 128, 512, 1024, 2048, 4096\}$ at n=100 per cell. The "killer comparison" compares *intact refusal at largest* N against *ablated refusal at* N=0: a small gap means dilution behaves mechanistically like full $\hat{d}$ removal.

Interpretation. The dilution effect is not a generic consequence of long context. It requires a prompt structure that produces *competing tasks* which pull Guardrail-Head attention away from

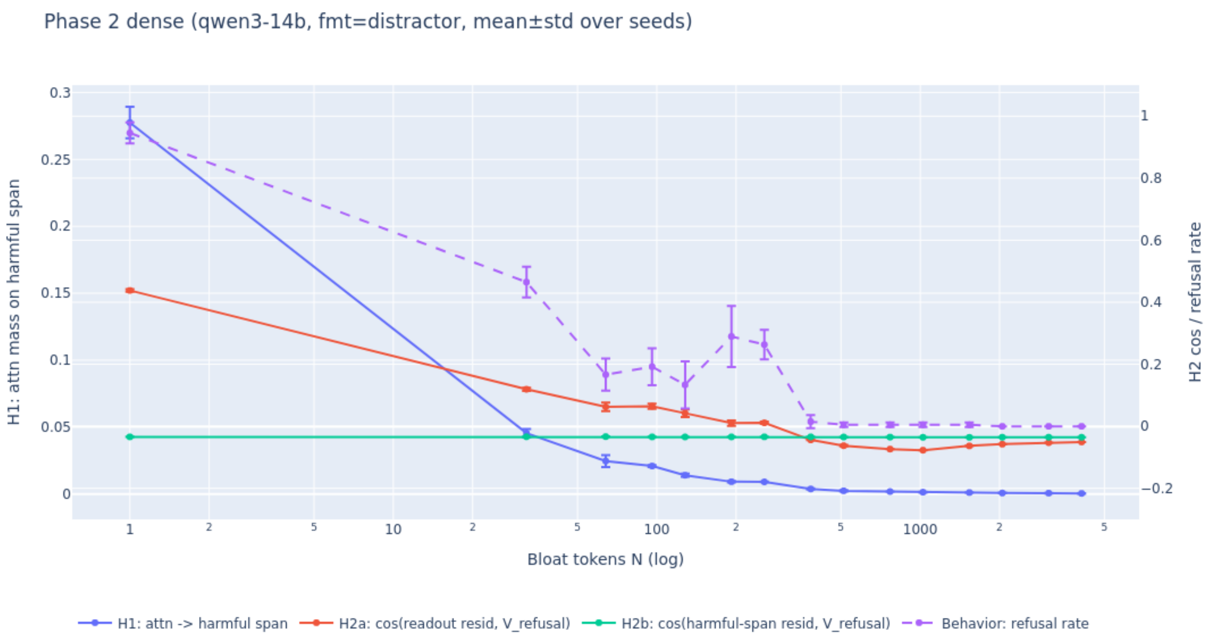


Figure 3: Phase 2 dense sweep on the focal `distractor` format across $N \in \{0, 32, 64, \ldots, 4096\}$. Three quantities overlaid: H1 = mean attention mass from L36's last token onto the harmful-token span at the Guardrail Heads (left axis); H2 = $\cos(\text{resid}_\ell^{(\text{last})}, \hat{d})$ at ℓ^*=36; and behavioral refusal rate. The behavioral curve tracks H1 closely; H2 stays close to constant. Half-baseline crossing at $N_{\text{jb}} \approx 31$ tokens. Red markers indicate runs that OOMed and are reported as failures rather than dropped.

the harmful tokens through softmax dilution. Inert prefix prose, conversational chat history, and few-shot Q/A blocks do not engage this mechanism, even at N=4096. This refines the popular framing of long-context jailbreaks: the relevant axis is structure, not raw token count.

8 Mechanism: Attentional, Not Representational

We measure the fraction of attention from L36's final-token position that the dominant Guardrail Head L36H31 places on the harmful-token span, over 24 held-out harmful prompts, comparing N=0 to N=4096 on the focal `distractor` format (Table 7). We use reduce-on-the-fly hooks (the hook computes the mass and discards the full attention pattern) to avoid OOM at long N.

By contrast, the magnitude of $\hat{d}$ in the residual stream is unchanged across N (the cosine in Table 5 is magnitude-normalized; we discuss the unnormalized signed projection in Section 9). Hypothesis H1 (attentional dilution) is supported; H2 (representational dilution) is falsified. The direction is intact; the head whose job it is to write that direction into the residual has stopped reading from the relevant tokens. This is consistent with Zhao et al. [2025]: harmfulness is encoded at the instruction tokens, refusal is computed downstream by reading from those tokens. Cut the read with softmax dilution and the downstream refusal computation never fires.

Format	intact @ N=4096	ablated @ N=0	gap
`distractor`	**0.00**	0.01	**0.01**
`suffix`	0.59	0.01	0.58
`sandwich`	0.87	0.01	0.86
`many_shot`	0.96	0.01	0.95
`prefix`	0.97	0.01	0.96
`multi_turn`	0.99	0.01	0.98

Table 6: Killer comparison per format: only `distractor` closes the gap to the ablation floor. Its dilution behavior is mechanistically equivalent to V_refusal removal.

Condition	Attn fraction (L36H31)
N=0	0.0061 ± 0.0035
N=4096	0.0010 ± 0.0029
ratio	$\approx 6\times$ drop

Table 7: Per-head attention mass on the harmful-token span at L36H31, n=24 prompts. The dominant Guardrail Head loses 6× of its attention mass on the harmful tokens under dilution.

9 The Failed Single-Site Rescue and the Sign-Flip Discovery

9.1 Activation steering does not rescue at high N

Following Turner et al. [2023], we install a single-site additive hook at L36's `resid_post` that adds $\alpha \cdot \hat{V}_{\text{refusal}}$ to every token position during inference, sweeping $\alpha \in \{0, 1, 2, 4, 8, 16\}$ at five N values on the focal `distractor` format (Table 8). At $N \geq 512$, no value of α restores refusal. At N=256 the rescue is weak ($0.23 \to 0.29$ as $\alpha : 0 \to 16$).

9.2 Capability-cost audit confirms $\hat{d}$ is narrow

A 2×2 capability check ($\{\text{intact, ablated}\} \times \{N{=}0, N{=}4096\}$) on 200 held-out MMLU items and 50 GSM8K items (Table 9) shows V_refusal removal costs only ∼2 pp on MMLU and induces zero spurious benign refusals. The direction is genuinely capability-orthogonal — so the failed rescue is not because we found the wrong direction.

9.3 Why the rescue fails: the $\hat{d}$ projection inverts sign

In the path-patching pre-computation (Section 10) we measure the unnormalized scalar projection of L36's last-token residual onto $\hat{d}$ over 8 held-out harmful prompts in clean (N=0) and corrupted (N=512 `distractor`) conditions:

	mean clean projection	mean dirty projection
across 8 prompts	+512	−66

Seven of eight prompts have their $\hat{d}$-projection flip sign under dilution. The model is not merely losing the refusal signal; it is actively producing a small *anti-refusal* signal. This quantitatively explains the failed rescue: at α=16 we add +16 units; the residual is at −66 in $\hat{d}$-units; restoring the projection to a positive value would require $\alpha \gtrsim 70$, an order of magnitude more than

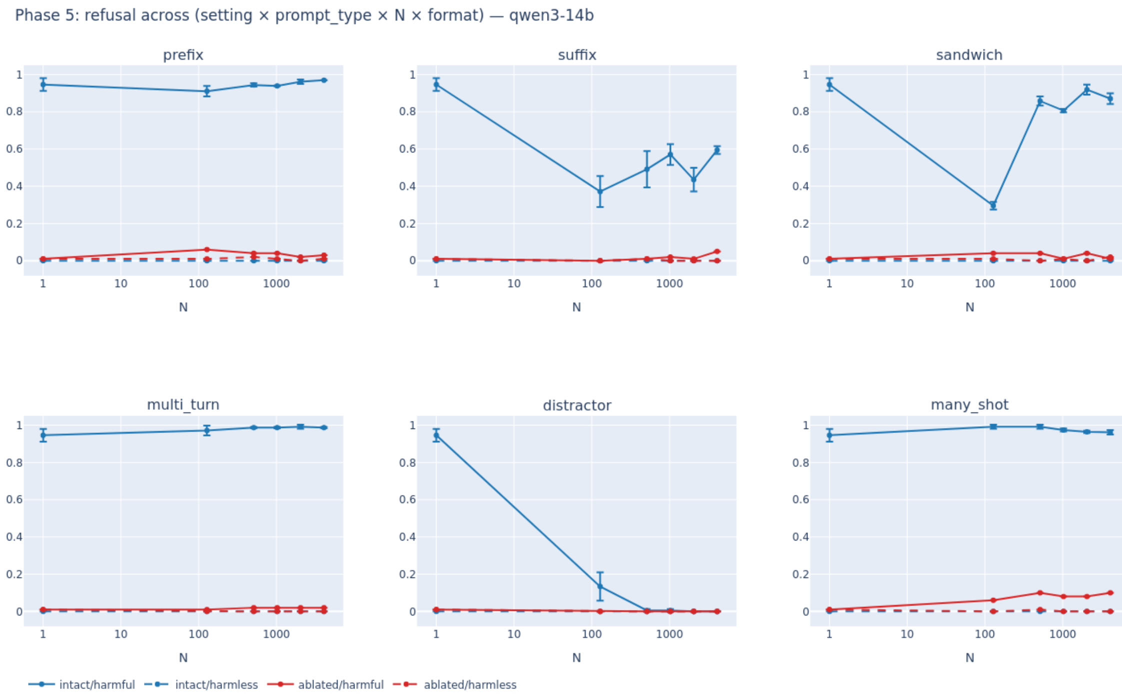


Figure 4: Six-panel cross-format 2×2 comparison ({intact, ablated} × {harmful, harmless}) across all six bloat formats (rows: format; columns: N on log axis). The intact-harmful curve (blue) collapses only on the `distractor` panel and overlays the ablated-harmful floor (red); in all five other formats it stays well above. Harmless curves (green/orange) sit at ~ 0 everywhere. This is the headline visual for "format > length": same model, same harmful prompts, same token budgets — only the wrapping structure varies.

tested. The cosine measurements in Table 5 stay close to zero rather than negative because they are magnitude-normalized over the full residual; the signed projection reveals that the V_refusal component is actively pushed into the negative half-space.

10 Path Patching Localizes the Lesion to a Two-Head Circuit

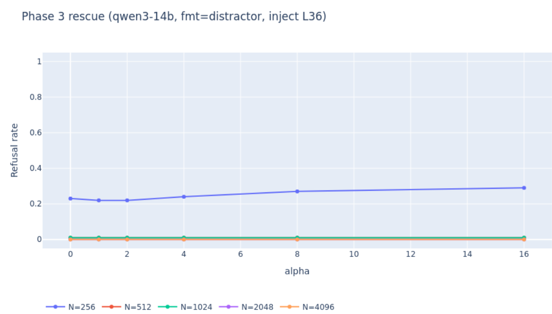
To localize the upstream cause of L36H31's attention collapse, we perform denoising path patching [Goldowsky-Dill et al., 2023] on 8 held-out harmful prompts. For each prompt, we cache the clean (N=0) residual stream at every layer, then run the corrupted (N=512 `distractor`) prompt while patching `resid_pre[`ℓ`]` at the harmful-span token positions from the clean cache. We report *recovery* as the fraction of the (clean−dirty) gap in the $\hat{d}$ projection at L36's last token restored by the patch: 0 means no effect; 1 means fully restored.

10.1 Layer sweep: peak at L=20

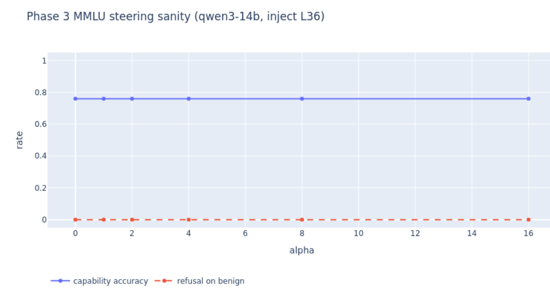
Table 10 reports per-layer recovery (mean ± SEM, n=8). A clean mid-network bump centered at **L=20**, with shoulders at L=16 and L=24, and recovery indistinguishable from zero at the embedding (ℓ<12) and at the readoff layer (ℓ≥28). Patching at L=36 itself has no effect — by then the damage is locked in. The localization is robust to leave-one-out: excluding a single high-recovery outlier

N	α=0	α=1	α=2	α=4	α=8	α=16
256	0.23	0.22	0.22	0.24	0.27	0.29
512	0.01	0.01	0.01	0.01	0.01	0.01
1024	0.01	0.01	0.01	0.01	0.01	0.01
2048	0.00	0.00	0.00	0.00	0.00	0.00
4096	0.00	0.00	0.00	0.00	0.00	0.00

Table 8: Refusal rate under single-site $\alpha \cdot \hat{d}$ injection at L36, focal format `distractor`, n=100 prompts per cell. Single-direction rescue fails for $N \geq 512$.



(a) Refusal rate vs. α on the focal format, one curve per N.



(b) MMLU accuracy and benign refusal vs. α on a 50-question subset.

Figure 5: (a) The single-site $\alpha \cdot \hat{d}$ rescue at L36 fails to restore refusal for $N \geq 512$ regardless of α. (b) The same intervention at the same α values does not damage benign capability (accuracy flat at 0.76, benign refusal at 0.00); the rescue lever is capability-safe but ineffective against dilution. Section 9 explains the failure as a sign flip in the $\hat{d}$ projection.

prompt, the L=20 mean falls to 6.6% $\pm$ 2.5% SEM (n=7), still maximal across the layer grid.

10.2 Joint patching reveals a coupled cascade

Joint patching at $\{16, 20, 24\}$ simultaneously yields recovery of 0.115 ± 0.041 SEM. This exceeds the best single layer (0.093 at L=20) but falls short of the sum of singles (0.178), indicating a **subadditive cascade**: clean residuals at upstream layers partially propagate downstream, so single-layer patches fix overlapping fractions of the same cascading lesion. This rules out "three independent circuits" — the three layers participate in a coupled mechanism (Figure 8).

10.3 Per-head zoom: L20H10 is uniquely dominant

We patch each head's `attn.hook_result` individually at $\ell \in \{16, 20, 24\}$ (Table 11). At L=20 a single head, **L20H10**, dominates (2.5× the next head). At L=16 the recovery is diffuse across many heads (top five all in 0.10–0.21%); at L=24 the recovery is essentially zero across all heads. L20H10 is therefore the unique upstream attention head whose silencing under dilution is causally responsible for L36H31's collapse on the harmful tokens.

10.4 The two-head mechanism

Combining all three path-patching results yields the mechanism:

Setting	N	MMLU acc.	GSM8K acc.	benign refusal
intact	0	0.760	0.02	0.000
intact	4096	0.702	0.08	0.000
ablated	0	0.737	0.04	0.000
ablated	4096	0.731	0.02	0.000

Table 9: Capability cost of V_refusal ablation on benign tasks. The direction is narrow: removing it costs $\sim$2 pp on MMLU and nothing on benign refusal.

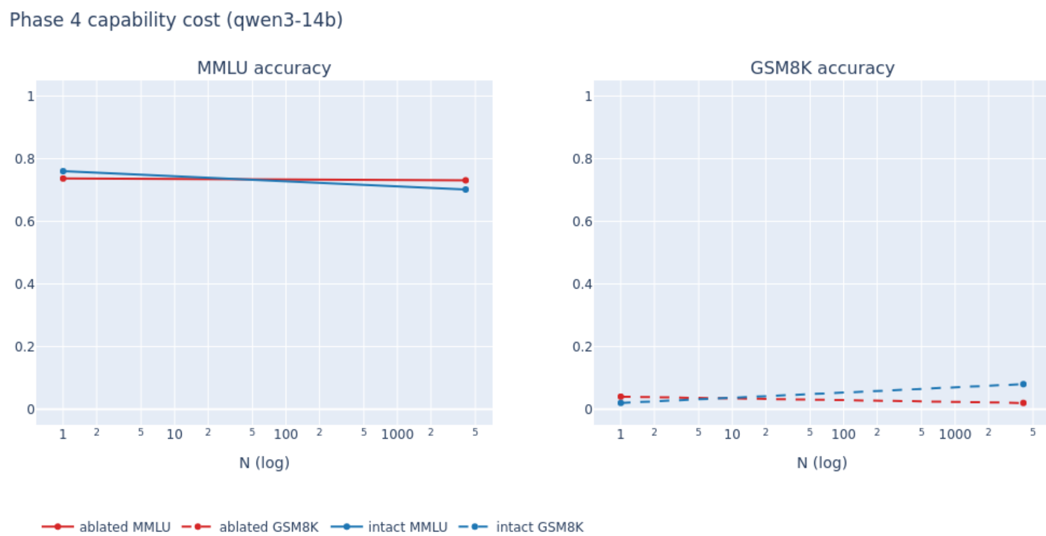


Figure 6: Phase 4 capability audit: MMLU and GSM8K accuracy under {intact, ablated} $\times$ $\{N{=}0, N{=}4096\}$. The ablated curve (red) tracks the intact curve (blue) within $\sim$2 pp on MMLU and is indistinguishable on GSM8K, supporting the claim that $\hat{d}$ is a narrow refusal-only direction rather than an entangled competence direction.

$$\textbf{L20H10} \rightarrow \text{(mid-layer cascade)} \rightarrow \textbf{L36H31} \rightarrow \hat{d} \text{ write at L36's last token}$$

Long-context `distractor` bloat silences L20H10's contribution at the harmful-token positions; the effect propagates through layers 16–24 (subadditively); L36H31 loses attention mass on the harmful tokens (the 6$\times$ drop in Table 7). L36H31's write to the residual stream is the canonical $\hat{d}$ direction; without that write, the residual at L36's last position drifts into the anti-refusal half-space (Section 9) and behavior collapses.

11 Discussion

The failure is more specific than "long context breaks refusal." The popular framing of long-context jailbreaks is that benign filler dilutes attention generally and the safety prior loses to bulk. Our results refine this in three ways. *(i) Format, not length.* Five of six bloat formats at N=4096 preserve refusal at $\geq$87%. Only `distractor` breaks the model, and it does so at $\sim$31 tokens. The relevant quantity is whether the prompt structure produces *competing tasks* that pull attention from the harmful tokens. *(ii) Local in depth.* The lesion is at a single mid-network band

Patched layer ℓ	Mean recovery	SEM
0	0.015	0.007
4	0.005	0.003
8	0.009	0.003
12	0.014	0.008
16	0.059	0.028
20	**0.093**	**0.034**
24	0.026	0.013
28	0.001	0.001
32	0.001	0.001
36	0.001	0.000

Table 10: Path-patching layer sweep: mean recovery of the $\hat{d}$-projection at L36's last token, after patching `resid_pre`[ℓ] at the harmful-span tokens from clean (N=0) into corrupted (N=512 `distractor`), n=8 held-out harmful prompts.

Layer	Top head (mean recovery ± SEM)	Notes
L=16	H20 (0.21% ± 0.15%)	flat: top 5 in 0.10–0.21% range
L=20	**H10 (0.37% ± 0.20%)**	2.5× next head
L=24	H10 (0.034% ± 0.016%)	negligible across all heads

Table 11: Per-head recovery at three layers near the L=20 lesion. L20H10 is uniquely dominant; no analogous head exists at L=16 or L=24.

(L=16–24, peak L=20). By L=36 the damage is irreversible; by L=12 the relevant computation has not yet happened. *(iii) Local in head space.* A single attention head, L20H10, accounts for the largest single-head contribution at the lesion layer; the mechanism is a two-head circuit, not a diffuse multi-component failure.

Why the single-direction rescue cannot work, and what could. The Phase-3 negative result is initially puzzling: $\hat{d}$ is a real, narrow, causal direction (Tables 2 and 9), so why doesn't adding it back rescue refusal? The sign-flip finding resolves this: the L36 residual at the last token does not merely lose its $\hat{d}$ component; it acquires a *negative* $\hat{d}$ component (mean projection -66 across prompts). Restoring the refusal signal would require α an order of magnitude larger than tested. Two viable interventions for future work follow: larger or position-targeted α with a fine-grained capability check; or **patching upstream at L20H10 directly** rather than at the L36 residual — targeting the cause rather than the symptom.

Implications for safety evaluation. If a single, structurally simple bloat format defeats refusal at $\sim$31 tokens, then short-context refusal benchmarks **systematically overstate the safety margin of deployed long-context models**. The gap is not narrow: `distractor`'s intact-refusal-at-N=4096 is 0.00, against the same prompts' baseline refusal of 0.97. We recommend that long-context safety evaluation explicitly include format-conditional dilution with at least the distractor structure at multiple N values ($\{32, 128, 512, 4096\}$), and report the killer-comparison gap as a single scalar of how attentionally fragile a model's refusal is.

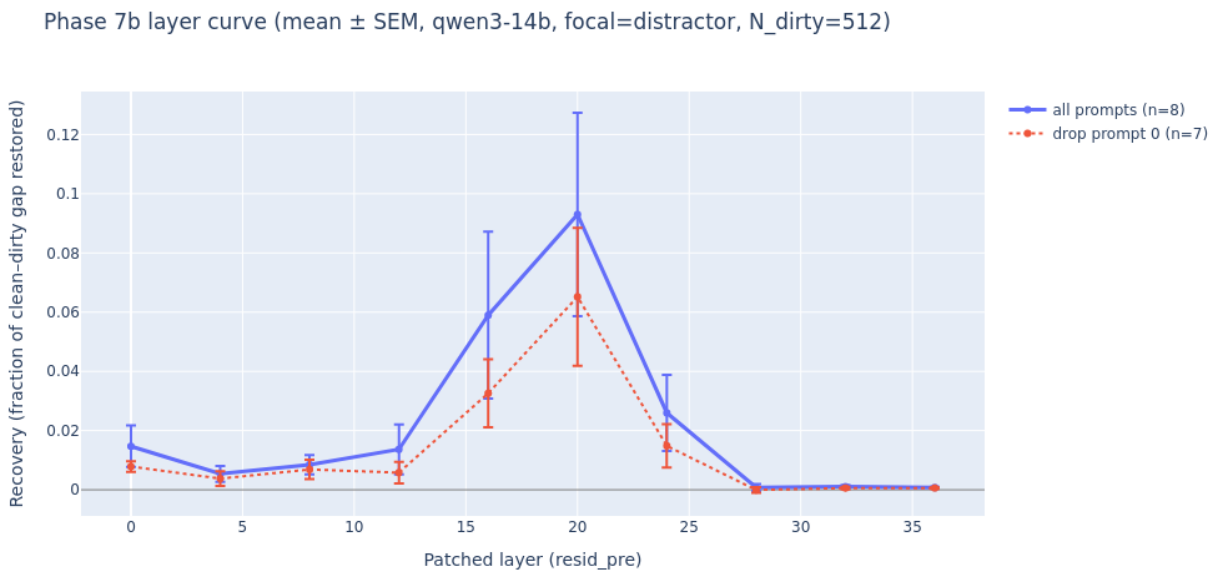


Figure 7: Phase 7b layer-sweep recovery curve, mean ± SEM (n=8). Solid line: all prompts. Dotted line: leave-one-out excluding the high-recovery outlier prompt 0 (n=7, mean L=20 recovery 6.6% ± 2.5%). Both curves peak at **L=20** with shoulders at L=16 and L=24, and recovery indistinguishable from zero at $\ell \geq 28$ (the readoff layer itself is the rightmost point and shows essentially no recovery — patching at L=36 cannot fix what is already wrong).

12 Limitations

- **One model.** All experiments are on Qwen3-14B. Arditi-style refusal directions transfer across model families, but the specific Guardrail Heads (L20H10, L36H31) and the exact dilution slope almost certainly do not. The two-head circuit claim needs replication on at least one Llama-family and one Mistral-family safety-tuned model.
- **Small prompt sets.** $\hat{d}$ is extracted from 256 paired prompts; Section 7 uses 100 per cell; path patching uses 8 per layer/head. The 8-prompt set is the limiting factor; joint and head-zoom recoveries have wide SEM bars.
- **Single bloat template per format.** Each of the six formats uses one canonical realization. Within-format variation in bloat content could shift slopes substantially.
- **Substring refusal detector.** Will mis-classify prefixed compliance ("I cannot help with that, but here is how…") as refusal and miss creative non-compliance. Model-judge or human-graded subset is needed before quoting the absolute refusal numbers as ground truth.
- **OOM-truncated long regime.** $N \geq 8192$ does not fit on a single A100-80GB with the current hook setup; longer-context rows are inferred indirectly via projection.
- **Three known overlap warnings** in `splits.json`; the harmless-mundane cell of the topic-decoupling test (Section 5.3) shares prompts with the $\hat{d}$ training pool (flagged, not corrected).
- **Path patching only at the harmful-span tokens.** Patching all positions would likely yield higher recovery and may shift the localization. The 9–12% recovery is therefore a *lower bound* on

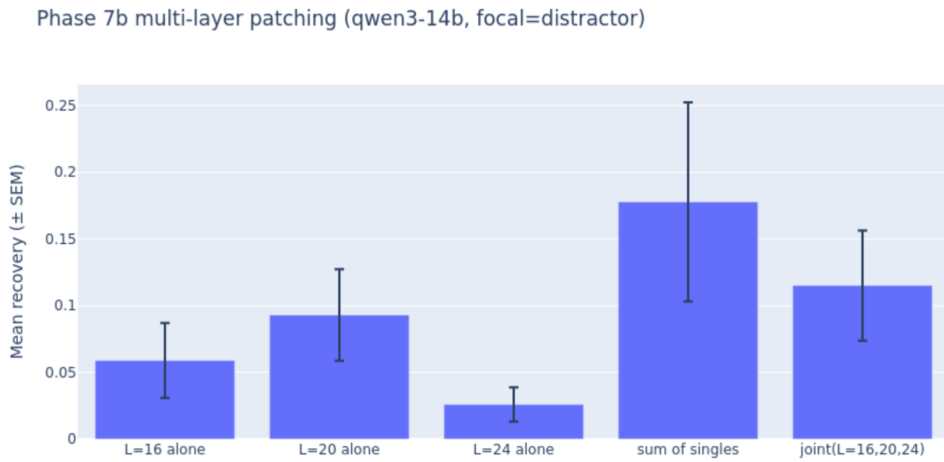


Figure 8: Multi-layer joint patching at $\{16, 20, 24\}$ vs. each single-layer patch and their naive sum. *Joint* $>$ *best single* (adding more layers helps marginally) but *joint* $<$ *sum of singles* (the layers act as a partially redundant cascade rather than independent contributors). Error bars are SEM, n=8.

the L=20 lesion's importance.

- **No mitigation evaluated.** Originally scoped LoRA-SFT plus DPO mitigation was descoped due to time; we discuss it as future work in Section 11.
- **Greedy decoding only.** All generation is `do_sample=False`; sampling-based jailbreak rates may differ.

13 Conclusion

We localized the long-context jailbreak failure mode in Qwen3-14B to a specific two-head circuit (**L20H10 → L36H31**) via a combination of difference-of-means refusal-direction extraction, six-format dilution sweeps, attention-mass measurement, capability-orthogonality auditing, and denoising path patching. The failure is **attentional, not representational**; it is **structure-conditional**, requiring task-competition formats like `distractor`; it produces a **sign flip** in the $\hat{d}$ projection that defeats single-site direction-level rescue; and it originates at a **single mid-network attention head** whose silencing cascades downstream to the dominant Guardrail Head at the readoff layer. Three concrete extensions follow: replication on Llama-family and Mistral-family safety-tuned models; feature-level rescue via attribution graphs (transcoder support permitting); and upstream intervention at L20H10 rather than at the L36 residual as a more effective steering target.

Author Contributions

Ayush Kokande led the four-experiment validity battery (Section 5) and the binary and continuous baseline N sweeps. Suraj Mishra led the $\hat{d}$ extraction at the canonical layer (Section 4.1), Guardrail-Head identification (Section 4.3), the six-format dilution analysis (Sections 6 and 7), attention-mass

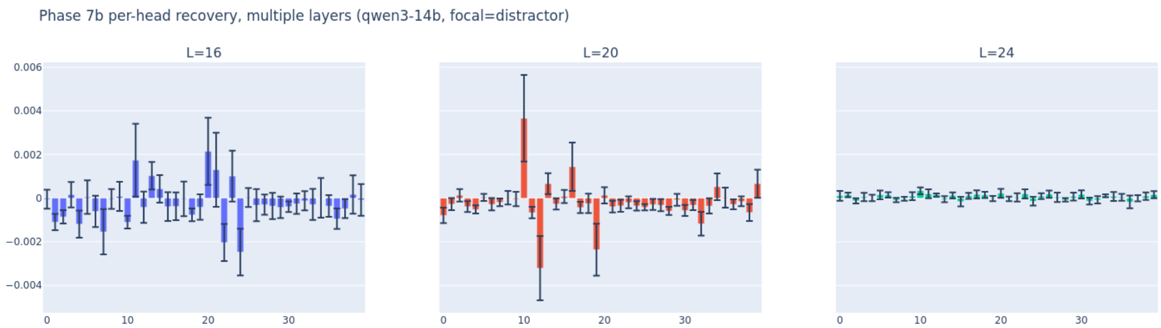


Figure 9: Per-head recovery (mean ± SEM, n=8) at the three layers around the L=20 lesion. **L=16**: flat distribution, no head dominates. **L=20**: head 10 stands out at 2.5× the next head, and is the only large-effect head at any patched (ℓ, h) cell tested. **L=24**: recovery is essentially zero across all heads. The L20H10 dominance is what allows us to compress the lesion description from "a mid-network circuit" to a two-head circuit (L20H10 → L36H31).

measurement (Section 8), the failed rescue and capability audit (Section 9), and path-patching localization (Section 10). Both authors contributed equally to writing.

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A Refusal Detector

The 18-marker substring list (lowercased, applied to the first 200 characters of the post-think-block completion): `"i cannot"`, `"i can't"`, `"i can not"`, `"i'm sorry"`, `"i am sorry"`, `"sorry"`, `"i'm not able"`, `"i am not able"`, `"i'm unable"`, `"i will not"`, `"i won't"`, `"cannot assist"`, `"can't assist"`, `"not able to"`, `"unable to assist"`, `"not appropriate"`, `"against my"`, `"i must decline"`. `strip_think_block` removes any leading `<think>`...`</think>` block from the raw completion before scoring.

B Splits and Disjointness

[TODO: Reproduce table from `splits.json`: per-experiment AdvBench and Alpaca row indices. Three known overlap warnings: (a) topic-decoupling harmless-mundane cell vs. $\hat{d}$ training pool, (b) [TODO: ...], (c) [TODO: ...].]

C Hyperparameters

Decoding. Greedy (`do_sample=False`, `temperature=0`), `max_new_tokens=24` for refusal generation, `max_new_tokens=8` for MMLU (single letter), 96 for GSM8K (chain-of-thought + numeric answer), 512 for the baseline Section 3.1.

$\hat{d}$ **extraction.** Training pools n=256 each; candidate layer grid for selection $\{14, 16, 18, \ldots, 36\}$ (integer step 2); validation pool n=100 each.

Ablation hooks. Direction projected out at `resid_pre`, `resid_mid`, `resid_post` of all 40 layers (120 hook sites total).

Steering hooks. Single-site additive hook at `blocks.36.hook_resid_post`; $\alpha \in \{0, 1, 2, 4, 8, 16\}$; $N \in \{256, 512, 1024, 2048, 4096\}$.

Path patching. n=8 harmful prompts; layer grid $\{0, 4, 8, \ldots, 36\}$; N_{dirty}=512 on focal `distractor`; reduce-on-the-fly attention hooks for the per-head zoom.

D Additional Figures

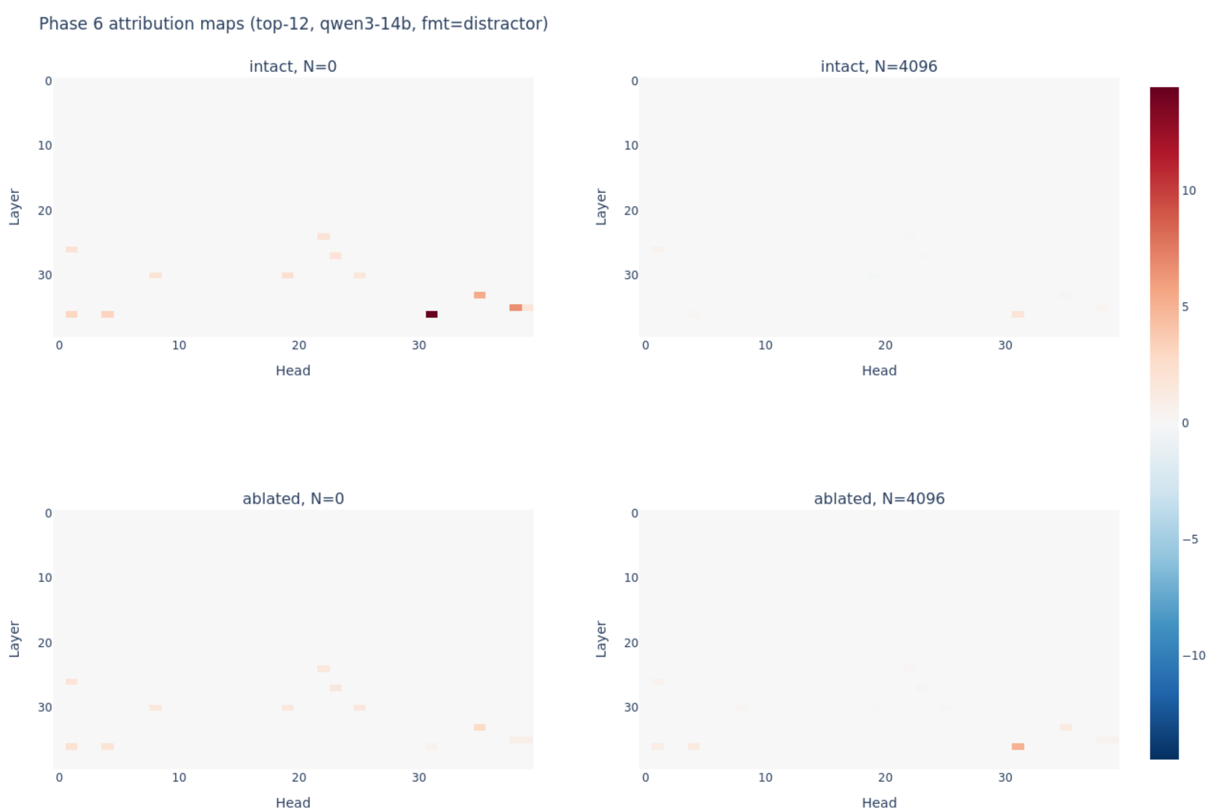

Figure 10: Phase 6 per-head attention attribution heatmap. Each cell shows the contribution of head (ℓ, h) to the V_refusal write at L36's last token under the focal `distractor` format. Generated using selective z-hooks on the top-K Guardrail Heads (reduce-on-the-fly to avoid materializing the full $[B, H, T, T]$ attention pattern at long N).

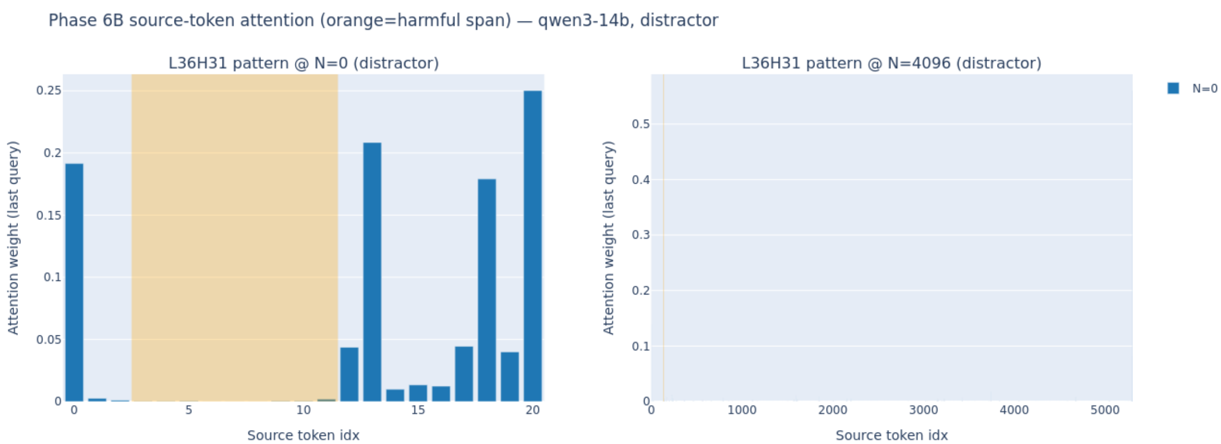

Figure 11: Phase 6 source-flow diagram: aggregated attention from L36H31's query at the final position to source-token positions across the prompt, comparing N=0 and N=4096. Under dilution the mass on the harmful-token span (highlighted) collapses by 6× relative to the no-bloat baseline; bloat positions absorb the redistributed attention.